Problem set 3:

Problem 1. Algebra of charges

Consider the Lagrangian

$$\mathcal{L} = \partial_\mu \Phi_a^\dagger \partial^\mu \Phi_a - V(\Phi_a)$$

Suppose it is symmetric under the

transf. $\phi^a \to \phi^a + i\theta_k (\hat{\lambda}_k)^a{}_b \phi^b + O(\theta^2)$

1) write down the conserved currents & conserved charges.

2) Use canonical commutation relations

$$[\pi_a(\vec{x}), \phi_b(\vec{y})] = -i\delta^d(\vec{x}-\vec{y})\,\delta_a{}^b$$

to compute the commutators

$$[J_0^a(\vec{x}), J_0^b(\vec{y})] = ?$$

$$[Q^a, J_0^b(\vec{x})] = ?$$

$$[Q^a, Q^b] = ?$$

Prob 2: Non-rel. Galilean Group

Consider the non-relativistic Galilean group generated by rotations & non-relativistic boosts $\vec{x}^i \to \vec{x}^i + \vec{v}^i t$

Show that this algebra can be centrally extended with the central charge that's the total mass of the system.

Prob 3: Adjoint representation

Show that $(\hat{\lambda}_k)^l{}_m = iC_{km}{}^l$ is a representation by checking the commutation relations for $(\hat{\lambda}_k)^l{}_m$

The resulting identity is called the Jacobi identity. Why does it hold?

What's the dimension of the Adj rep.?